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WADP 24x-3.4TiB NVMe 4xNDR200 Nclient1

System Description

Items	Descriptions
System Name	WADP 24x-3.4TiB NVMe 4xNDR200 Nclient1
Submitter Name	ExponTech
Submission Date	July 11, 2025
Benchmark Version	2.0
Solution Type	Remote Block
Submission Division	Closed
Training Workload	Resnet50,llama3-8b,llama3-70b,llama3-405b,llama3-1t

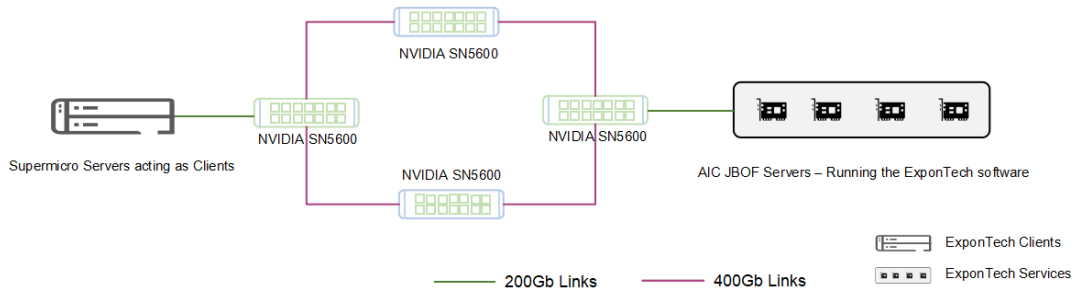
Note:

Subset is used for llama3 checkpointing, suchas llama3-8b, llama3-70b, llama3-405b, llama3-1t.

1 Hardware

1.1 System Topology

The client and the storage server are each connected to separate NVIDIA SN5600 switches. Both operate at 200 Gb/s, while the switches are interconnected at 400 Gb/s



1.2 Client

Client Description

- Client Quantity: 1
- Product model: Supermicro AS-4125GS-TNRT2
- OS Boot Drive: 1TB SATA HDD (Model: MM1000GBKAL)
- Client NIC Type: ConnectX-7 BlueField®-3 BF3140
- Client Number of Connected NIC Ports: 4
- Rack Unit: 2U * 1

1.3 Server

Equipment:

- Storage Servers Description: 1 AIC JBOD F2026, internally configured with 4 BlueField®-3 BF3220SH
- Storage Servers OS Boot Drive : 480GB SSD
- Storage SSD: 24 * 3.84TB NVMEs
- NVMe SSD Model: ScaleFlux CSD5000-CSDU7UXF38
- Storage Servers NIC Type : ConnectX-7 BlueField®-3 BF3220SH
- Storage Servers Number of Connected NIC Ports: 4
- Rack Unit: 2U * 1

Power Requirement:

System component	Quantity	Power supply unit	Nameplate rated power	Design power
Server	1	Power supply 1	3200 watts	3200 watts
		Power supply 2	3200 watts	
Total			6400 watts	3200 watts

1.4 Switch

The switch is a commercial NVIDIA SN5600 (920-9N42F-00RI-7C0). The client and Storage server are connected on it.

2 Software

2.1 Client

Client Operating System:

- Client OS : ubuntu 24.04 LTS
- Client Kernel : 6.8.0-62- generic

DPU Operating System:

- Client DPU OS : ubuntu 22.04.5 LTS
- Client DPU Kernel : 5.15.0-1060-bluefield

2.2 Server

- ExponTech WADP software : ET-WADP-3.6.1
- Storage Servers OS: ubuntu 22.04.5 LTS
- Storage Servers Kernel : 5.15.0-1060-bluefield

2.3 Switch

- Switch OS: Cumulus Linux 5.12.1
- Switch Kernel: 6.1.0-cl-1-amd64